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Simple Algebra

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: Solve: 4(x + 3) = 2x + 22.
Method & answer: 4x + 12 = 2x + 22, so 2x = 10, x = 5.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Solve: 7x − 6 = 43.

2. Solve: 5x + 8 = 4x + 19.

3. Solve: 3(x + 5) = 33. A) 6 B) 7 C) 8 D) 9

4. Solve: 3x/4 = 9.

5. Solve: 6(x − 2) = 4x + 6.

6. Select ALL of the equations below where x = 7: A) 5x − 4 = 31 B) 2(x + 2) = 18 C) 4x + 3 = 31 D) 3x − 2 = 19 E) 6x − 1 = 40

7. Solve: 8x + 2 = 6x + 20.

8. Solve: (x − 4)/3 = 5.

9. Solve: 5(2x − 1) = 55. A) 5 B) 6 C) 7 D) 8

10. Two numbers have a sum of 44 and a difference of 6. Find the larger number.

11. Solve: 10x − 3 = 7x + 18.

12. Select ALL of these equations where x = 8: A) 4x − 5 = 27 B) 3x + 4 = 28 C) x/4 + 3 = 5 D) 6x − 10 = 38 E) 5x + 1 = 50

13. Solve: 7(x + 2) − 3x = 38.

14. Solve: 4x + 7 = 3x + 7 + 9. A) 6 B) 7 C) 8 D) 9

15. Solve: 12x − 4(x + 1) = 44.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Solve: 2(x + 4) = 3(x − 2).

2. Solve: (3x + 6)/3 = 9.

3. Solve: 4(x − 3) = 2(x + 5). A) 9 B) 10 C) 11 D) 12

4. The sum of three consecutive numbers is 84. Find the largest number.

6. Select ALL of the equations below where x = 9: A) 4x − 6 = 30 B) 3(x + 1) = 30 C) 6x − 15 = 39 D) 5x + 2 = 47 E) 7x − 3 = 65

7. A number is trebled, then 9 is subtracted, giving 36. Find the number.

8. Solve: 4(2x + 1) − 6 = 46.

9. Two numbers have a sum of 72 and one is 3 times the other. Which is the larger number? A) 18 B) 36

11. Select ALL of these equations where x = 6: A) 4(x+1) = 28 B) 3x + 12 = 30 C) 5x − 8 = 22 D) (x+6)/2 = 6 E) 7x − 5 = 40

12. A rectangle's length is 3 times its width. If the perimeter is 64cm, find the width.

13. x satisfies 4x + 5 = 53. Find x.

14. Solve: 9x − 3(x + 2) = 42. A) 6 B) 7 C) 8 D) 9

C) 54 D) 60

5. Solve: 5x + 2(x − 3) = 43.

10. Solve: (x + 8)/5 = x − 4.

15. A number doubled and increased by 13 equals the number trebled decreased by 2. Find the number.

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Solve: 6(x − 4) = 3(x + 5).

6. A father's age is 5 times his son's age. In 8 years, he will be 3 times his son's age. How old is the son now?

11. A taxi fare is £5 plus £2.20 per mile. A second taxi has a flat fare of £22.60. After how many miles do the two fares become equal?

2. Two numbers have a sum of 96 and a difference of 22. Which is the smaller number? A) 33 B) 35 C) 37 D) 39

7. Solve: 9x − 4(x − 3) = 3x + 32. A) 8 B) 9 C) 10 D) 11

12. The sum of two numbers is 100. One number is 16 more than 3 times the other. Find the larger number.

3. A number squared is 15 more than 2 times the number. Find the positive value of the number.

8. The perimeter of an isosceles triangle is 46cm. The two equal sides are each 5cm longer than the base. Find the base length.

13. Solve: 9x − 5(x − 3) = 2x + 39. A) 10 B) 11 C) 12 D) 13

4. Select ALL of the equations below where x = 11: A) 5x − 12 = 43 B) 3(x − 4) = 21 C) 2x + 7 = 30 D) 4x − 9 = 35 E) 6x − 5 = 61

9. Two consecutive odd numbers multiply to give 195. Find the larger number.

14. A number multiplied by itself gives the same result as 6 times the number plus 27. Find the positive value of the number.

5. A rectangle's length is 5cm more than twice its width, and the perimeter is 70cm. Find the width.

10. Select ALL of the values of x below that make 4x − 9 greater than or equal to 27: A) 8 B) 9 C) 10 D) 11 E) 12

15. Two numbers have a ratio of 3:5 and their sum is 96. Find the difference between the two numbers.